

Testing Validation substeps

Validate the design through planned tests, evidence capture, defect review, and closure

1. Define validation scope and acceptance criteria

Clarify what this validation cycle is proving, which requirements or risks it covers, and what success looks like. Include pass/fail limits, expected behavior, and any exceptions that are acceptable.[[emmainternational](#)]

2. Build the validation test plan and traceability

Create the test plan and map each test to the requirement, function, or risk it supports. This makes it clear why each test exists and what evidence it must produce.[[elib.dlr](#)]

3. Prepare the test environment and fixtures

Set up the hardware, software, tools, samples, fixtures, and safety checks needed to run the tests properly. Confirm the environment matches the intended use as closely as practical.[[pdsvision](#)]

4. Execute validation tests and capture evidence

Run the planned tests and record the measured results, observations, logs, screenshots, and other proof of performance. Keep the execution consistent so the data is reliable and repeatable.[[safetyculture](#)]

5. Review failures, root causes, and retest needs

Compare the results against acceptance criteria, identify any failures or unexpected behavior, and decide whether the issue is a test problem, design problem, or requirement issue. If needed, define what must be retested after corrections.[[ops.fhwa.dot](#)]

6. Close corrective actions and approve validation

Assign action owners, track fixes, verify closure, and document the final approval decision. This step ensures the validation effort ends with a clear sign-off and traceable evidence.[[fip.r-a-w](#)]